

PRACTICAL-7

AIM:

Wap to create a string like, "Hello World" and perform some operations.

- (A) To perform all indexing / slicing operations.
- (B) To perform update operation in string.
- (C) To perform upper case and lower case operations.
- (D) To perform delete operations.

CODE:

```
#Creating the string
text = "Hello World"
# Indexing / Slicing operations
print("Indexing Operations:")
print("First character:", text[0])
print("Last character:", text[-1])
print("Characters from index 0 to 4:", text[0:5])
print("Characters from index 6 to end:", text[6:])
#Update operation in string
updatedtext = text[:6] + "Universe"
print("\nUpdated Text:", updatedtext)
#Upper case and lower case operations
print("Upper case:", text.upper())
print("Lower case:", text.lower())
# Delete operations
text_after_delete = text.replace("World", "")
print("\nAfter deleting 'World':", text_after_delete)
```

OUTPUT:

```
Indexing Operations:
First character: H
Last character: d
Characters from index 0 to 4: Hello
Characters from index 6 to end: World

Updated Text: Hello Universe

Upper case: HELLO WORLD
Lower case: hello world

After deleting 'World': Hello
```

PRACTICAL-8

AIM:

Wap to create the list like [1,2,3, "Rohit","Ankit",True] and perform some operations.

- (A) To perform all indexing/slicing operations in above list.
- (B) To perform all update operations in above list.
- (C) To perform all Delete operation in above list.
- (D) To perform all reverse operations in above list.

CODE:

```
# Creating the list
my_list = [1, 2, 3, "Rohit", "Ankit", True]
# (A) Indexing / Slicing operations
print("Indexing Operations:")
print("First element:", my_list[0])
print("Last element:", my_list[-1])
print("Elements from index 1 to 3:", my_list[1:4])
print("Elements from index 2 to the end:", my_list[2:])
# (B) Update operations in the list
print("\nUpdate Operations:")
my_list[1] = 10
my_list[5] = False
print("Updated List:", my_list)
# (C) Delete operations in the list
print("\nDelete Operations:")
del my_list[2]
print("List after deletion at index 2:", my_list)
my_list.remove("Rohit")
print("List after removing 'Rohit':", my_list)
my_list.pop()
print("List after popping the last element:", my_list)
# (D) Reverse operations on the list
print("\nReverse Operations:")
my_list.reverse()
print("Reversed List:", my_list)
```

OUTPUT:

Indexing Operations:

First element: 1

Last element: True

Elements from index 1 to 3: [2, 3, 'Rohit']

Elements from index 2 to the end: [3, 'Rohit', 'Ankit', True]

Update Operations:

Updated List: [1, 10, 3, 'Rohit', 'Ankit', False]

Delete Operations:

List after deletion at index 2: [1, 10, 'Rohit', 'Ankit', False]

List after removing 'Rohit': [1, 10, 'Ankit', False]

List after popping the last element: [1, 10, 'Ankit']

Reverse Operations:

Reversed List: ['Ankit', 10, 1]

PRACTICAL-9

AIM:

Wap to Perform the Sum, Len, Avg, Min, Max Method in the list.

CODE:

```
# Creating the list
my_list = [1, 2, 3, 4, 5]
# (A) Sum of the list
sum_of_list = sum(my_list)
print("Sum of the list:", sum_of_list)
# (B) Length of the list
len_of_list = len(my_list)
print("Length of the list:", len_of_list)
# (C) Average of the list
avg_of_list = sum_of_list / len_of_list
print("Average of the list:", avg_of_list)
# (D) Minimum value in the list
min_of_list = min(my_list)
print("Minimum value in the list:", min_of_list)
# (E) Maximum value in the list
max_of_list = max(my_list)
print("Maximum value in the list:", max_of_list)
```

OUTPUT:

```
Sum of the list: 15
Length of the list: 5
Average of the list: 3.0
Minimum value in the list: 1
Maximum value in the list: 5
```

PRACTICAL-10

AIM:

Wap to perform all possible Operations in Tuple.

CODE:

```
tuple1 = (1, 2, 3, 4, 5)
```

```
tuple2 = ('a', 'b', 'c')
```

```
#Access elements by index
```

```
print("Accessing elements by index:")
```

```
print("First element of tuple1:", tuple1[0])
```

```
print("Last element of tuple2:", tuple2[-1])
```

```
#Slice a tuple
```

```
print("\nSlicing a tuple:")
```

```
print("Slice from index 1 to 3 of tuple1:", tuple1[1:4])
```

```
print("Slice of all elements in tuple2:", tuple2[:])
```

```
#Concatenate two tuples
```

```
print("\nConcatenating tuples:")
```

```
tuple3 = tuple1 + tuple2
```

```
print("Concatenated tuple:", tuple3)
```

```
#Repeat a tuple
```

```
print("\nRepeating a tuple:")
```

```
tuple4 = tuple1 * 2
```

```
print("Repeated tuple1:", tuple4)
```

```
#Get length of a tuple
```

```
print("\nLength of tuple1:", len(tuple1))
```

```
print("Length of tuple2:", len(tuple2))
```

OUTPUT:

```
Accessing elements by index:
First element of tuple1: 1
Last element of tuple2: c

Slicing a tuple:
Slice from index 1 to 3 of tuple1: (2, 3, 4)
Slice of all elements in tuple2: ('a', 'b', 'c')

Concatenating tuples:
Concatenated tuple: (1, 2, 3, 4, 5, 'a', 'b', 'c')

Repeating a tuple:
Repeated tuple1: (1, 2, 3, 4, 5, 1, 2, 3, 4, 5)

Length of tuple1: 5
Length of tuple2: 3
PS C:\Users\pawan\OneDrive\Desktop\python>
```

PRACTICAL-11

AIM:

Wap to Perform all possible Operations in Dictionary.

CODE:

```
dict1 = {'name': 'John', 'age': 25, 'city': 'New York'}
dict2 = {'course': 'Python', 'duration': '3 months'}
# Access values using keys
print("Name:", dict1['name'])
# Add a new key-value pair
dict1['email'] = 'john@example.com'
print("Updated dict1:", dict1)
# Modify an existing key's value
dict1['age'] = 26
print("Updated dict1 with modified age:", dict1)
# using `del`
del dict1['city']
print("dict1 after removing 'city':", dict1)
# using `pop()`
removed_value = dict2.pop('duration')
print("Removed 'duration':", removed_value)
print("Updated dict2:", dict2)
# Get the value of a key using `get()`
print("Get value of 'age':", dict1.get('age'))
print("Get value of 'email':", dict1.get('email'))
# Get all keys or values using `keys()` and `values()`
print("Keys of dict1:", dict1.keys())
print("Values of dict1:", dict1.values())
# length of a dictionary
print("Length of dict1:", len(dict1))
# Clear all items from the dictionary
dict1.clear()
print("dict1 after clearing all items:", dict1)
```

OUTPUT:

```
dict1 = {'name': 'John', 'age': 25, 'city': 'New York'}
dict2 = {'course': 'Python', 'duration': '3 months'}

# Access values using keys
print("Name:", dict1['name'])

# Add a new key-value pair
dict1['email'] = 'john@example.com'
print("Updated dict1:", dict1)

# Modify an existing key's value
dict1['age'] = 26
print("Updated dict1 with modified age:", dict1)

# using `del`
del dict1['city']
print("dict1 after removing 'city':", dict1)

# using `pop()`
removed_value = dict2.pop('duration')
print("Removed 'duration':", removed_value)
print("Updated dict2:", dict2)

# Get the value of a key using `get()`
print("Get value of 'age':", dict1.get('age'))
print("Get value of 'email':", dict1.get('email'))

# Get all keys or values using `keys()` and `values()`
print("Keys of dict1:", dict1.keys())
print("Values of dict1:", dict1.values())

# length of a dictionary
print("Length of dict1:", len(dict1))

# Clear all items from the dictionary
dict1.clear()
print("dict1 after clearing all items:", dict1)
```